

DA 2 Progress Report: 50% Completion

Project: Intelligent Router Path Selection using Machine Learning

Repository: <https://github.com/K-RANGARAJAN/smart-path-selector> (tag da2)

1. Summary

The data and modelling half of the project is complete. A multi-AS network simulator produces time-varying traffic with congestion episodes. RIP, OSPF and BGP path selection run on the same topology. A 280,000-row dataset has been generated, and a Random Forest predicts next-interval path quality on unseen traffic with **$R^2 = 0.897$ and MAE = 4.7 quality points** (0-100 scale).

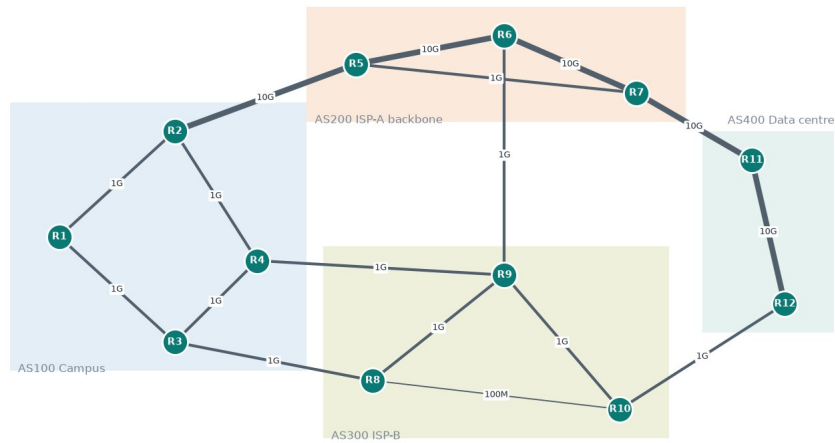
Before any ML routing, the baseline analysis already shows the gap this project targets. On unseen traffic the static protocols deliver a mean path quality of 50-90, while the best available path each tick would deliver about 96.

2. Completion status

Component	Status	Notes
Problem definition, architecture	Done	DA 1
Topology (12 routers, 4 ASes, 17 links)	Done	smartpath/topology.py
Traffic and congestion model	Done	smartpath/traffic.py
Link/path metrics and quality score	Done	smartpath/metrics.py
RIP / OSPF / BGP decision logic	Done	smartpath/protocols.py, tested against Dijkstra
Candidate path enumeration	Done	smartpath/paths.py
Dataset generation	Done	python -m smartpath generate
Random Forest training and evaluation	Done	python -m smartpath train
Unit tests	Done (16 passing)	pytest
Recommendation engine	Remaining	Final review
Closed-loop routing evaluation vs protocols	Remaining	Final review
Live dashboard	Remaining	Final review
Packet Tracer validation	Remaining	Final review

3. What was built

3.1 Topology



AS	Role	Characteristics
AS100	Campus (sources R1-R4)	1 Gbps internal links
AS200	ISP-A backbone	10 Gbps, heavily shared, frequent congestion
AS300	ISP-B	1 Gbps links plus a 100 Mbps shortcut (R8-R10)
AS400	Data centre (destinations R11, R12)	10 Gbps

The topology is built so that each protocol makes a different, defensible choice.

Flow	RIP (hop count)	OSPF (cost)	BGP (policy)
R1→R12	R1-R3-R8-R10-R12 (4 hops, crosses 100 Mbps)	R1-R2-R5-R6-R7-R11-R12 (cost 15)	R1-R2-R4-R9-R10-R12 (via preferred AS300)
R1→R11	R1-R2-R5-R7-R11	R1-R2-R5-R6-R7-R11	R1-R2-R4-R9-R10-R12-R11
R2→R12	R2-R4-R9-R10-R12	R2-R5-R6-R7-R11-R12	R2-R4-R9-R10-R12
R3→R11	R3-R8-R10-R12-R11	R3-R1-R2-R5-R6-R7-R11	R3-R4-R9-R10-R12-R11

3.2 Traffic model

Each link's utilisation is a mean-reverting AR(1) process around a base load. On top of that sit a daily cycle and randomly arriving congestion episodes (10-60 ticks, +25-60% utilisation, ramping up and fading). Episodes can also be injected by hand for demonstrations.

3.3 Metrics and quality score

Per link, from utilisation u :

- queueing delay $\propto u/(1-u)$ (M/M/1 shape, capped at 250 ms)
- jitter $\propto$ queueing delay
- loss: 0.05% floor, rising sharply as the buffer overflows near $u \approx 0.9$
- available bandwidth = capacity $\times (1-u)$

Path metrics aggregate across links: latency sums, jitter adds in quadrature, loss compounds, and available bandwidth is the bottleneck.

Quality score (0-100) = simplified ITU-T G.107 E-model R-factor (delay, jitter, loss impairments) $\times \sqrt{\text{bandwidth sufficiency for a 150 Mbps flow}}$.

Routers observe paths through **noisy probes**: $\pm 8\%$ latency noise, $\pm 25\%$ jitter noise, $\pm 10\%$ bandwidth noise, and loss counted over 100 pings. The model has to predict the **true** quality in the **next** tick from these measurements.

3.4 Dataset

	Train	Test
Traffic seed	1	2 (independent traffic)
Ticks	6,000	2,000
Rows (tick $\times$ flow $\times$ candidate path)	210,000	70,000
Flows	4	4
Candidate paths per flow	8-9	8-9

Features (9): latency, jitter, packet loss, available bandwidth, hop count, bottleneck capacity, and the change in latency, loss and bandwidth since the previous measurement. Target: realised quality score in the next tick. Mean 73.0, median 96.6. 26.5% of samples are below 60 (visibly degraded).

4. Model results

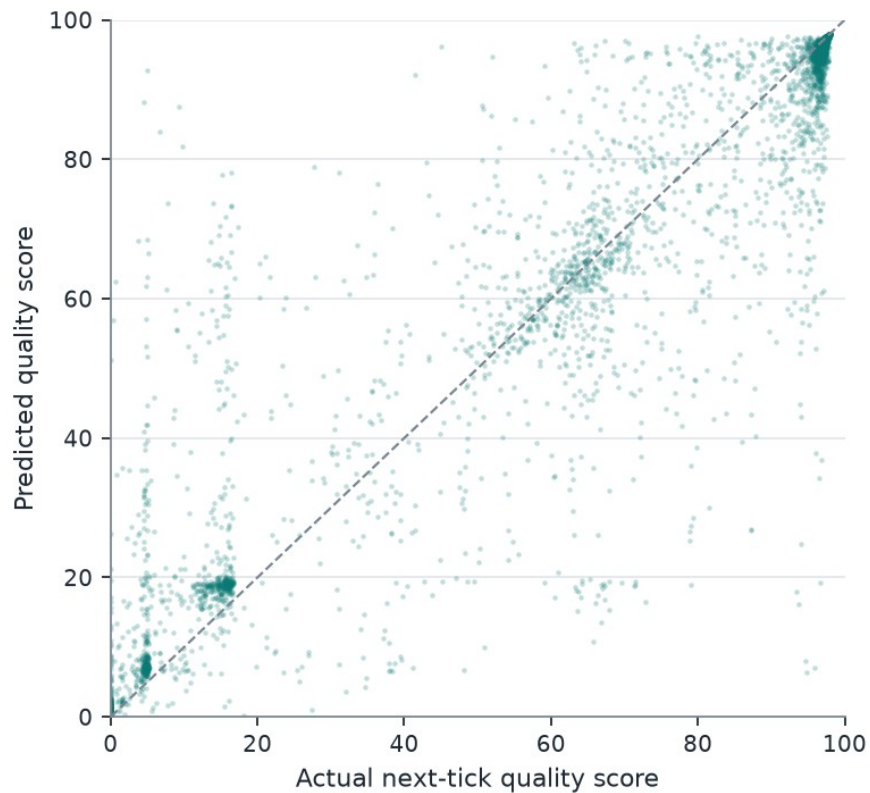
Random Forest: 150 trees, max depth 16, min 20 samples per leaf, 60% of features per split. Training takes 14 s. Inference is about 0.002 ms per path (batched).

4.1 Accuracy on unseen traffic (test seed)

Predictor	R ²	MAE	RMSE
Random Forest	0.897	4.69	10.77
Scoring formula on current measurements (no ML)	0.876	5.56	11.82
Linear regression	0.786	10.30	15.53

4.2 Cross-validation (5 folds, contiguous time blocks)

R² = **0.912 $\pm$ 0.009**, MAE = 4.50

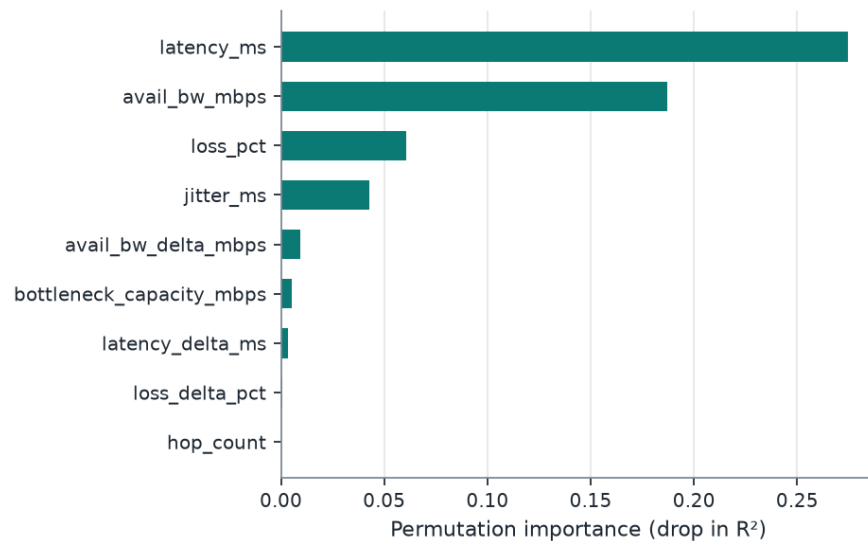


Most predictions sit on the diagonal. The large errors happen when a path's state changes sharply between the measurement and the next tick, for example when a congestion episode starts or ends. The absolute error correlates at $r = 0.92$ with the size of that change.

4.3 What drives the prediction (permutation importance on test data)

Feature	Drop in R^2 when shuffled
Latency	0.275
Available bandwidth	0.187
Packet loss	0.061
Jitter	0.043
Bandwidth trend	0.009
Bottleneck capacity	0.005
Latency trend	0.003
Loss trend	0.001
Hop count	0.0001

Hop count, the only metric RIP uses, carries essentially no information about how a path will perform.



5. Baseline analysis: static protocols on unseen traffic

Mean next-tick quality of each protocol's fixed path, compared with the best fixed path in hindsight and with the per-tick best path (oracle):

Flow	RIP	OSPF	BGP	Best fixed path	Oracle	Best-path changes
R1→R11	89.5	75.4	73.1	89.5	95.8	161
R1→R12	50.1	74.9	73.4	89.0	96.1	177
R2→R12	73.8	75.5	73.8	89.6	96.0	165
R3→R11	50.0	75.1	73.3	89.4	96.4	395

Observations:

- No single fixed path is best. The best path changes 160-400 times in 2,000 ticks. A static protocol cannot follow this, however well it is configured.
- OSPF puts traffic on the 10 Gbps backbone, which is the most congested part of the network.
- RIP's result depends on the topology. For R1→R11 the fewest-hop route avoids the congested backbone and does well (89.5). For R1→R12 and R3→R11 it crosses the 100 Mbps link and scores about 50.
- The gap from the static protocols to the oracle (about 6-46 points) is what the ML engine aims to close.

6. Issues found and how they were resolved

Issue	Resolution
In the first calibration the backbone was congested almost all the time, so one static path was simply best and there was nothing to learn	Retuned base loads so the backbone is usually good but has congestion bursts; the best path now changes regularly
BGP initially chose the same paths as OSPF	Set local preference to favour ISP-B (the cheaper contract), so BGP follows policy rather than speed, which is realistic
First model file was 200 MB	Limited tree depth and leaf size: 16 MB, with slightly better test accuracy (R^2 0.896 → 0.897)
The measured-metrics formula is already a strong predictor (R^2 0.876)	Kept it as a named baseline so ML gains are reported honestly rather than only against static protocols

7. Remaining work (to final review)

1. Recommendation engine: score candidates each tick, add hysteresis against route flapping.
2. Closed-loop evaluation: ML, a non-ML adaptive router, RIP, OSPF, BGP and the oracle route the same flows on a fresh traffic seed. Compare quality, latency, loss, time degraded and route changes.
3. Live dashboard: topology with link load, candidate path scores and each strategy's choice, with congestion injection for the demo.
4. Packet Tracer validation: parse ping/tracert output and score measured paths.

8. Reproducing these results

```
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python -m smartpath generate # datasets, topology figure, baseline tables
python -m smartpath train   # model, metrics JSON, figures
pytest
```